Fraudulent Claim Detection System

Leveraging Machine Learning for Enhanced Risk Management

Problem Statement

- **Current Challenge:**
  - Manual fraud detection process
  - Leads to delays and increased payouts
  - Significant financial risks for Global Insure

- **Business Goal:**
  - Develop a machine learning-based fraud detection system
  - Classify claims as fraudulent or legitimate
  - Enable early intervention
  - Reduce financial losses and optimize claims handling

Methodology Overview

- **Structured Approach:**
  - Data Understanding
  - Data Cleaning & Preprocessing
  - Train-Validation Split
  - Exploratory Data Analysis (EDA)
  - Feature Engineering
  - Model Building
  - Performance Evaluation

Data Understanding & Cleaning

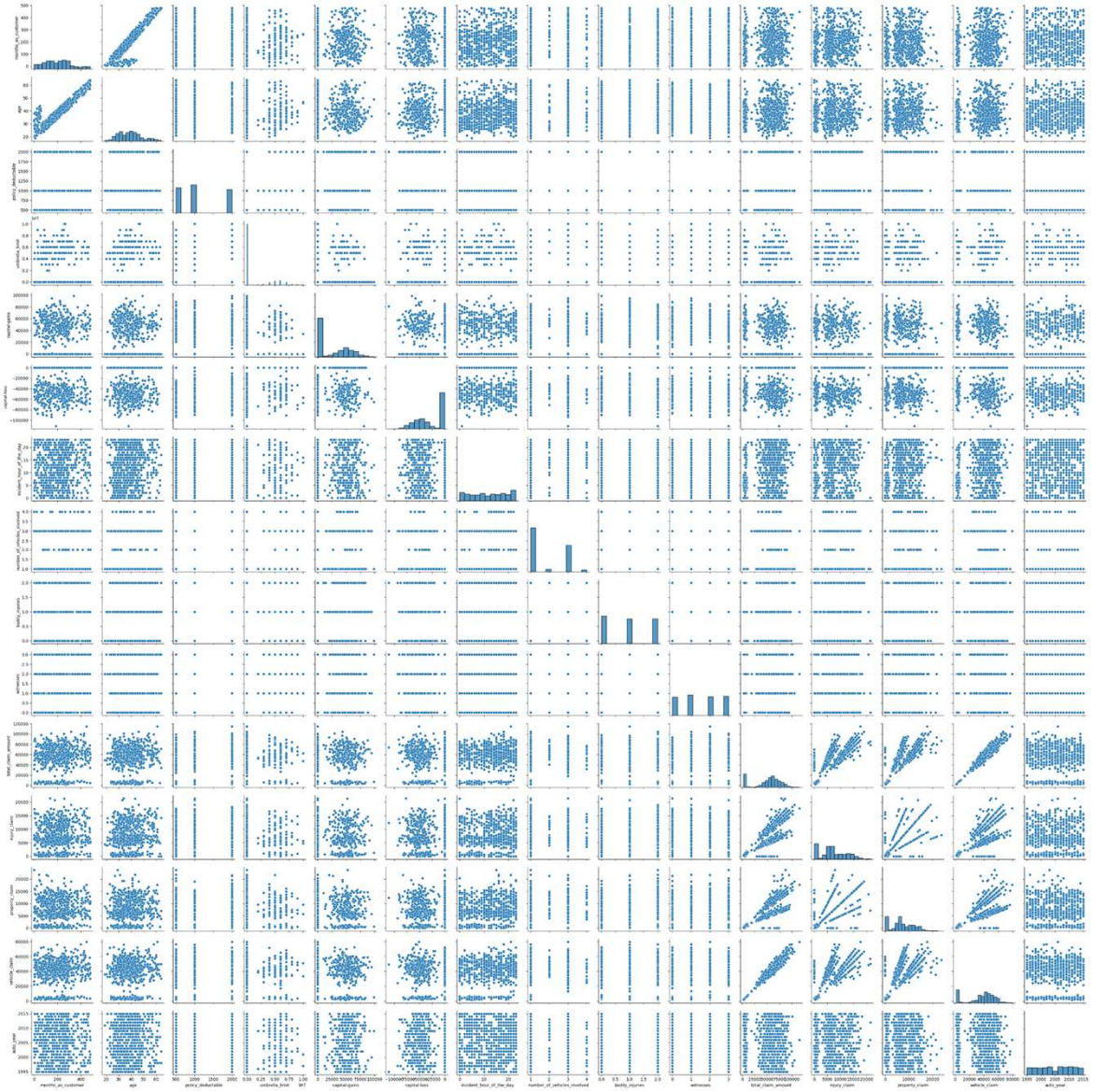
- **Data Understanding:**
  - Analyzed claim data: customer info, policy details, claim amounts, incident descriptions, fraud reports.
- **Data Cleaning:**
  - Remove irrelevant columns
  - Handle missing values
  - Identify and handle redundant/illogical values
  - Fix data types

Exploratory Data Analysis (EDA)

Univariate and Bivariate Analysis

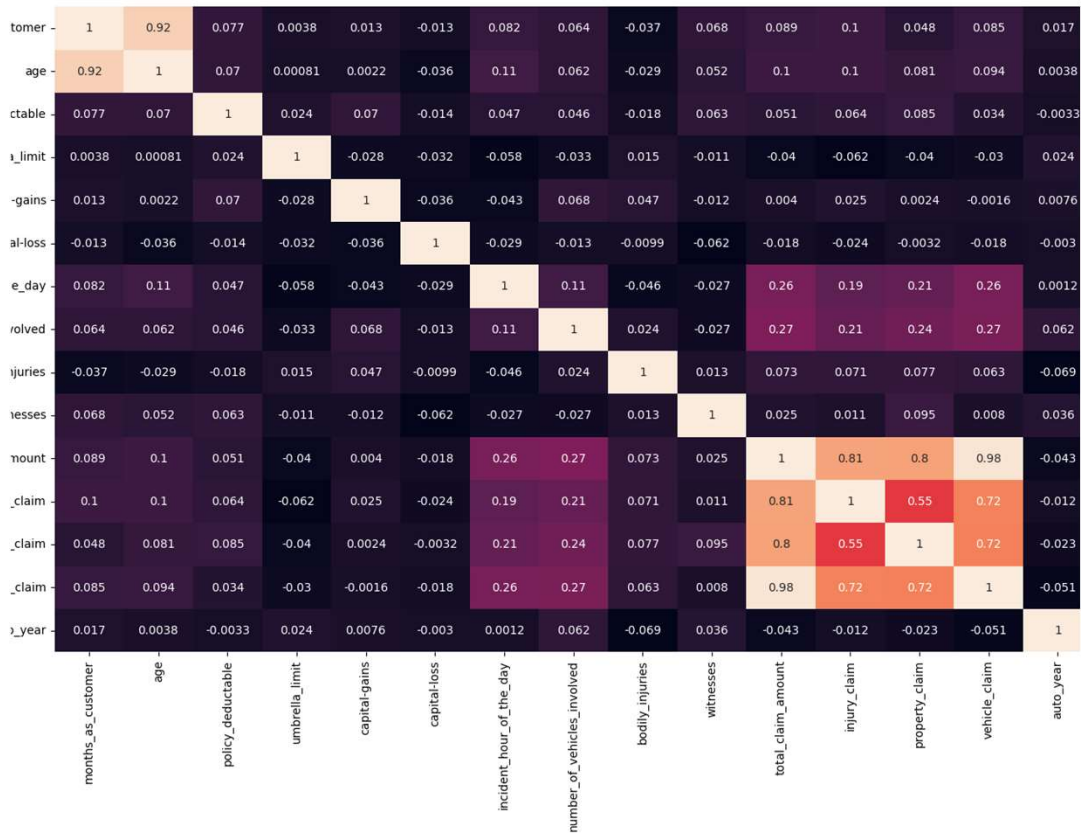
Numerical Distribution

Visualized distributions of numerical columns



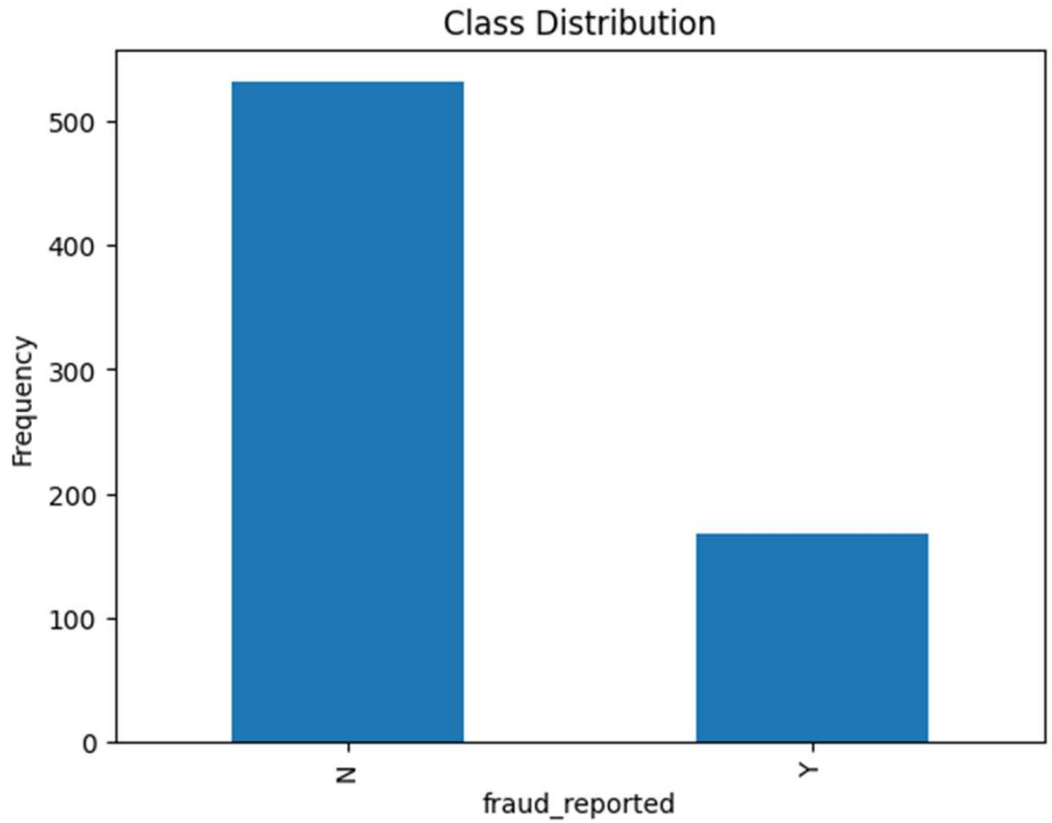
Correlation Analysis

Heatmap revealed high correlations between financial variables (e.g., claim amounts), requiring careful feature selection.



Class Imbalance

Target variable (fraud_reported) is highly imbalanced (approx. 24% fraudulent claims).



Bivariate Analysis

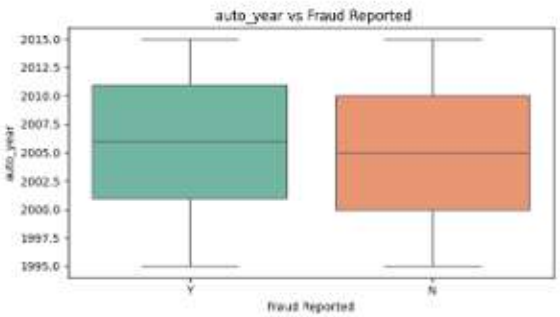
Examined relationships between categorical columns and target.

Feature: incident_type

incident_type	Count	Fraud_Likelihood
Multi-vehicle Collision	419	0.272076
Parked Car	84	0.095238
Single Vehicle Collision	402	0.291045
Vehicle Theft	94	0.085106

Feature: incident_severity

incident_severity	Count	Fraud_Likelihood
Major Damage	275	0.607273
Minor Damage	354	0.107345
Total Loss	280	0.128571
Trivial Damage	90	0.066667



Feature Engineering

- Techniques Applied:
  - Oversampling: Used RandomOverSampler to balance the imbalanced dataset.
  - New Feature Creation: Developed features based on domain understanding.
  - Redundant Column Dropping: Eliminated columns from which necessary features were extracted.
  - Dummy Variables: Transformed categorical variables into numerical representations.Standardization:
  - Applied StandardScaler to numerical columns for faster model convergence and balanced feature weights.

Model Building

- **Models Developed:**
  - Logistic Regression
  - Random Forest Classifier

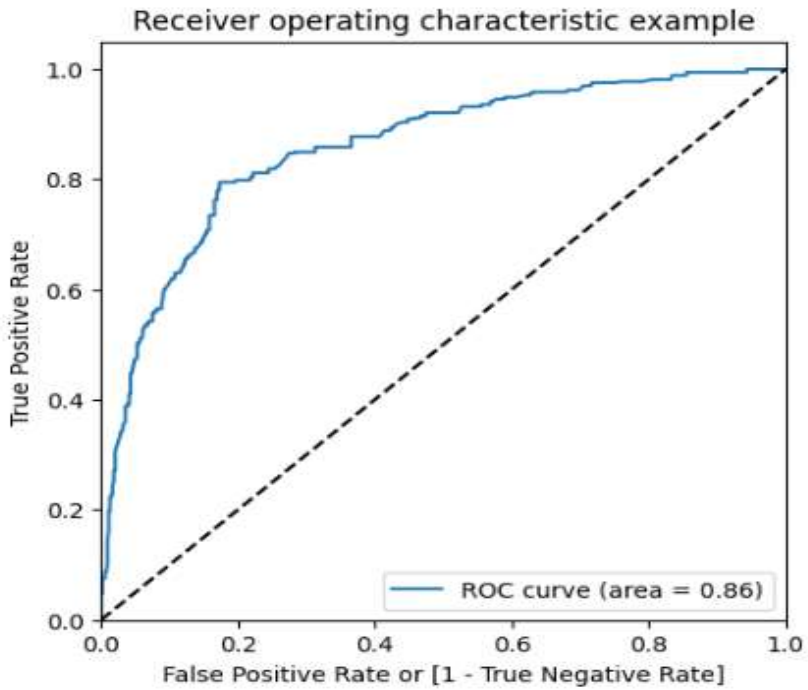
- **Process:**
  - **Feature Selection:** Applied Recursive Feature Elimination with Cross-Validation (RFECV).
  - **Multicollinearity Check (Logistic Regression):** Assessed using p-values and VIF.
  - **Training & Optimization:** Trained both models on training data and optimized hyperparameters through validation.

Model Evaluation & Cutoff Optimization

- Evaluation Metrics:
  - Precision
  - Recall (with focus on fraud detection)
  - F1-Score (for balanced assessment)
- Cutoff Optimization:
  - Created a cutoff optimization plot to balance recall, F1-score, sensitivity, and specificity at different probability thresholds.
  - Optimal cutoff identified at 0.48.

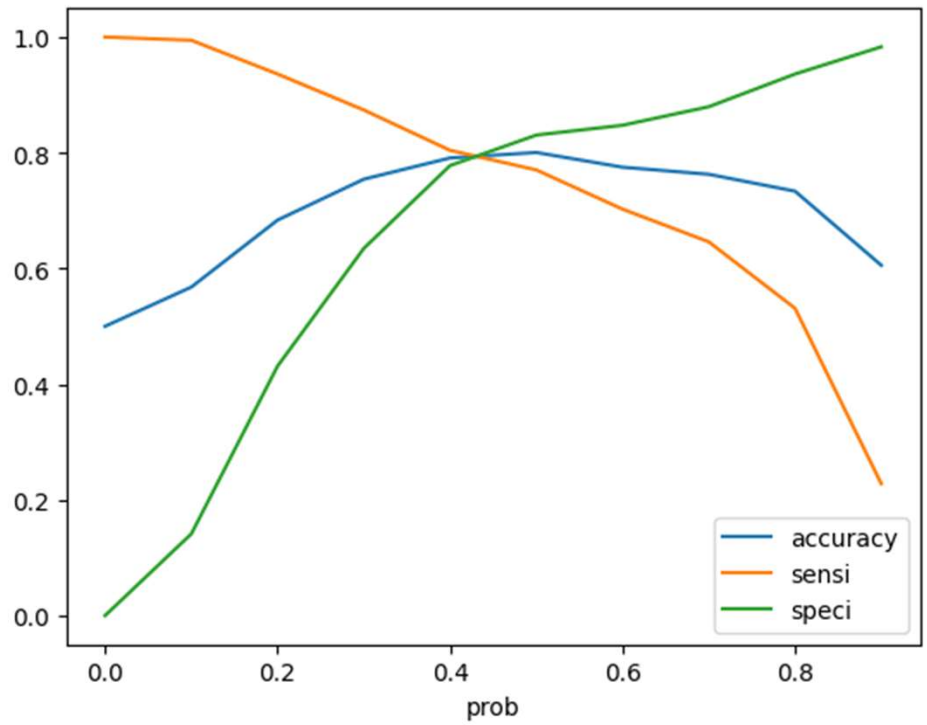
Visualizations

- ROC Curve



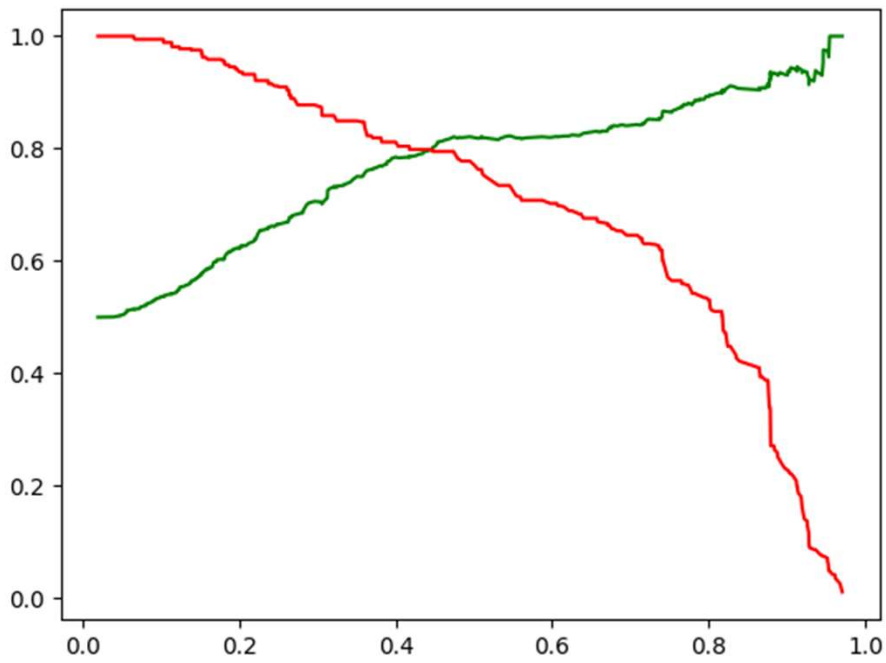
Visualizations

- Accuracy, Sensitivity, Specificity vs. Cutoff Plot



Visualizations

- Precision-Recall Curve



Sensitivity, Specificity, Precision, Recall and F1-score

- Logistic Regression Model

Training Data		
	Sensitivity	0.78
	Specificity	0.83
	Precision	0.82
	Recall	0.78
	F1	0.80
Validation Data		
	Sensitivity	0.68
	Specificity	0.79
	Precision	0.54
	Recall	0.68
	F1	0.60

- Random Forest Model

Training Data		
	Sensitivity	0.74
	Specificity	0.85
	Precision	0.83
	Recall	0.74
	F1	0.78
Validation Data		
	Sensitivity	0.65
	Specificity	0.83
	Precision	0.57
	Recall	0.65
	F1	0.61

Model Results & Recommendation

- Performance Comparison:

MODEL	PRECISION	RECALL (Fraud)	F1 SCORE
Logistic Regression	0.54	0.68	0.60
Random Forest	0.57	0.65	0.61

Recommendation

- **Random Forest** achieves better precision and F1-score.
- Offers a more balanced performance between fraud detection and false alarms.
- Recommended for deployment, especially where missing fraud is more costly than investigating false positives.

Assumptions & Limitations

- Assumptions:
  - Redundant columns were dropped.
  - Dummy variables were created.
  - Optimal cutoff is 0.48.
- Limitations:
  - Real-world fraud rates may differ, potentially impacting precision.
  - Assumes feature availability at claim submission for real-time predictions.
  - Current approach focuses only on structured data, not text or images.

Q&A

Q) How can we analyse historical claim data to detect patterns that indicate fraudulent claims?

A) Global Insure can analyze historical claim data to detect fraud patterns through a comprehensive machine learning methodology:

- Data Understanding & Cleaning: This initial phase involves analyzing various data points such as customer information, policy details, claim amounts, incident descriptions, and fraud reports. Key steps include removing irrelevant columns, handling missing values, identifying and correcting redundant or illogical entries, and fixing data types.
- Exploratory Data Analysis (EDA): EDA utilizes univariate and bivariate analysis, visualizing numerical data distributions, and performing correlation analysis. A heatmap can reveal strong correlations between financial variables like different claim amounts, necessitating careful feature selection. EDA also highlights class imbalance, where approximately 24% of claims are fraudulent.
- Feature Engineering: Techniques such as oversampling (using RandomOverSampler) are applied to address dataset imbalance. New features are created based on domain understanding, redundant columns are removed, and dummy variables are used to convert categorical data into numerical formats. Numerical columns undergo standardization with StandardScaler for faster model convergence and balanced feature weighting.
- Model Building: Logistic Regression and Random Forest Classifier models are developed. This stage includes feature selection using Recursive Feature Elimination with Cross-Validation (RFECV), and checking for multicollinearity in Logistic Regression using p-values and VIF. Both models are trained on the data and their hyperparameters are optimized.
- Performance Evaluation: Models are assessed using metrics like precision, recall (with a focus on fraud detection), and F1-score. An optimal cutoff probability of 0.48 is identified to balance these metrics. Visualizations like ROC curves and Accuracy, Sensitivity, Specificity vs. Cutoff plots aid in this evaluation.

Q) Which features are the most predictive of fraudulent behaviour?

A) Financial variables such as total_claim_amount, injury_claim, property_claim, and vehicle_claim as highly correlated. Bivariate analysis also shows that incident_type (e.g., Multi-vehicle Collision, Single Vehicle Collision) and incident_severity (e.g., Major Damage, Total Loss) are related to fraud likelihood. The feature engineering process further emphasizes the creation of new features based on domain expertise, suggesting that these engineered features, alongside selected original ones, are crucial for prediction.

Q) Based on past data, can we predict the likelihood of fraud for an incoming claim?

A) Yes, Global Insure can predict the likelihood of fraud for an incoming claim based on past data. The core business goal is to build a machine learning system to classify claims as fraudulent or legitimate. The developed models, specifically the Random Forest Classifier, can achieve this by outputting a probability score for each new claim, with an optimal cutoff of 0.48 used to flag claims for further investigation.

Q) What insights can be drawn from the model that can help in improving the fraud detection process?

A) The insights drawn from the model can significantly improve the fraud detection process:

- Proactive Fraud Detection: The system enables early intervention in claims handling, reducing potential financial losses that result from manual detection processes and delays.
- Optimized Resource Allocation: By classifying claims, the model allows for the prioritization of investigations, directing human fraud analysts to high-risk claims more efficiently.
- Balanced Performance: The Random Forest model is recommended for its superior precision and F1-score, providing a more balanced approach between detecting actual fraudulent claims and minimizing false alarms. This is particularly beneficial when the cost of missing fraud is higher than the cost of investigating a false positive.